

Selecting The RIGHT PCB BOARD MATERIALS

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Electronic devices, microwaves and other household devices rely on PCB technology to stay in working condition

Lifetime and performance of a PCB board depend on the choice of circuit board material. To select the right circuit board material, it is important to examine the materials available for different board categories. There are different properties and physical characteristics that help in the selection of board materials. While evaluating PCB material, it is necessary to ensure that design requirements align with desired board capabilities.

Spacing and width of a board are also important, when it is required to handle high amounts of current. Structural strength of the board is defined by substrate and laminate. Selection of materials for these two layers is based on the type of board.

Layers in a PCB board and their significance

A PCB board consists of four layers, namely, substrate, laminate, solder mask and silkscreen. Together, the substrate and laminate define the basic electrical, mechanical and thermal circuit board properties.

Substrate. Fibre-glass FR4 is the most common material for the substrate of a PCB. Here, FR stands for fire retardant. It is suitable due to its rigidity and thickness. For flexible PCB, Kapton or equivalent property plastic is used.

Thickness of the PCB board depends on its application or use. For example, most Sparkfun products are 1.6mm thick, while Arduino Pro ones use 0.8mm thick boards. A PCB built with less expensive materials like epoxies lacks durability.

Substrates are found in low-cost consumer electronics. These have low thermal stability, which causes them to lose their lamination easily. Substrates can also cause smoke when a soldering iron is held for long on the board, which makes them easy to identify.

A non-conductive layer of the dielectric material is chosen based on dielectric constant.

Substrates must meet certain desired properties like glass transition temperature (Tg). Tg is the point where heat causes the material to deform or soften. There is a wide variety of materials available for substrates, such as aluminium or insulated metal substrate (IMS) FR-1 through FR-6,

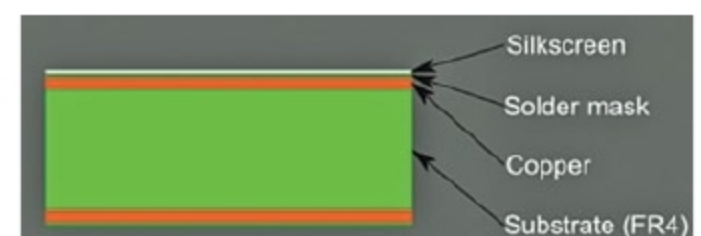


Fig. 1: PCB layer composition (Credit: <https://learn.sparkfun.com>)